

ShopEZ E-Commerce Shopping Platform

- Solution Requirements

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Team ID: TEAM-2026-001

Project Name: ShopEZ E-Commerce Shopping Platform

Functional Requirements

FR No.	Functional Requirement	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through email and password, password encryption using bcrypt.js, email verification, and profile creation.
FR-2	User Login	Login using email and password, JWT authentication, secure session management, and logout functionality.
FR-3	Product Catalog Management	View product listings, product details, categories, product images, and featured products.
FR-4	Product Search & Filtering	Search products by name, category, brand, or price range with advanced filtering options.
FR-5	Shopping Cart Management	Add products to cart, update quantity, remove items, and calculate total order value.
FR-6	Secure Checkout & Orders	Place orders securely, manage shipping details, generate invoices, and send order confirmation notifications.
FR-7	Order Tracking	View order history, track order status, delivery updates, and cancellation requests.
FR-8	Wishlist Management	Save products for future purchases and move items

		between wishlist and cart.
FR-9	Product Reviews & Ratings	Submit reviews, rate products, and view customer feedback.
FR-10	Admin Dashboard	Manage products, categories, inventory, users, orders, discounts, and promotions.
FR-11	Inventory Management	Track stock availability, update inventory levels, and receive low-stock alerts.
FR-12	Analytics & Reporting	View sales reports, revenue analytics, customer activity reports, and product performance statistics.
FR-13	Role-Based Access Control	Separate User and Admin roles with protected routes and authorization middleware.

Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Responsive and user-friendly interface with intuitive navigation across desktop, tablet, and mobile devices.
NFR-2	Security	JWT authentication, bcrypt password hashing, HTTPS communication, input validation, and secure API endpoints.
NFR-3	Reliability	Consistent order processing, data integrity, centralized error handling, and secure database transactions.
NFR-4	Performance	Product pages and APIs should respond within 2 seconds under normal load conditions.
NFR-5	Availability	Cloud deployment with 99.5% uptime and continuous service availability.
NFR-6	Scalability	MERN architecture capable of supporting increasing

		users, products, and transactions through horizontal scaling.
NFR-7	Maintainability	Modular code structure, reusable components, and proper documentation.
NFR-8	Compatibility	Support for modern web browsers including Chrome, Firefox, Edge, and Safari.

Conclusion

The ShopEZ E-Commerce Shopping Platform requires secure user authentication, efficient product management, seamless shopping functionality, and robust administration features. These requirements ensure a secure, scalable, reliable, and user-friendly platform.